ORIGINAL ARTICLE

Artificial Intelligence in Surgical Feedback: A Systematic Scoping Review

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OBJECTIVE: Artificial intelligence (AI) is poised to transform surgical education, particularly in providing objective, scalable feedback. This study aims to systematically evaluate the current landscape of AI applications in delivering procedural feedback to surgical learners.

METHODS: A systematic scoping review of literature published between 2015 and 2025 was conducted using search terms related to AI, surgical education, and feedback. Following PRISMA guidelines, studies were included if they detailed the implementation of an AI-based system for surgical or procedural feedback. Retrieved articles underwent title, abstract, and full-text screening to determine final eligibility.

RESULTS: The initial search yielded 2374 unique studies, with 18 meeting the final inclusion criteria. A risk of bias analysis revealed that twelve of these studies (67%) presented a moderate risk of bias. The included studies spanned multiple specialties, including general surgery (33%), neurosurgery (17%), and otolaryngology (6%). The efficacy of AI feedback was primarily assessed through performance metrics against expert benchmarks and subjective learner surveys. Analysis revealed that AI-driven feedback was, in several studies, found to be equivalent or superior to expert human instruction in simulation settings. While most studies (n = 14) utilized post-task feedback, real-time AI models demonstrated the potential for faster error correction. The majority of studies focused on trainees (n = 10), though AI was shown to be a valuable educational tool for learners at all skill levels. Five studies were randomized controlled trials, indicating a trend toward more rigorous study designs.

CONCLUSION: AI represents a promising and significant supplement to traditional procedural feedback methods. Future development, particularly for applications in otolaryngology, will depend on interdisciplinary collaboration to overcome barriers such as cost and user trust. (J Surg Ed 83:103908. © 2026 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

KEY WORDS: artificial intelligence, surgical education, feedback, machine learning, simulation, procedural skill

ABBREVIATIONS: AI, artificial intelligence; SBT, simulation-based training; VR, virtual reality; PRISMA-ScR, preferred reporting items for systematic reviews and meta-analyses extension for scoping reviews; RCT, randomized controlled trial; ROBINS-I, risk of bias in non-randomized studies of interventions; SVM, support vector machine; CNN, convolutional neural network; RNN, recurrent neural network; NLP, natural language processing; LLM, large language model; XAI, explainable AI; VOA, virtual operative assistant; ADR, adenoma detection rate; AUPRC, area under the precision-recall curve; CS, checklist score; CVS, critical view of safety; EASE, end-to-end assessment of suturing expertise; GPC, gastric precancerous conditions; GRS, global rating scale; GS, global score; ICEMS, intelligent continuous expertise monitoring system; LCRF, laparoscopic cholecystectomy rating form; OSATS, objective structured assessment of technical skills; RARP, robot-assisted radical prostatectomy; TWIX, training with explanations

INTRODUCTION

Artificial intelligence (AI) is rapidly emerging as a transformative force in medicine, with surgical education standing as a particularly promising frontier for its application. Procedural teaching, in particular, is well-suited for AI applications, given the critical need for unbiased assessment and feedback to guide skill acquisition. AI, broadly defined as the field of computer science that aims to simulate human intelligence, offers a transformative approach to traditional educational challenges.[1] The conventional

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model of surgical training is fundamentally limited by its reliance on one-on-one expert instruction, which is naturally subjective and difficult to scale. This creates a critical need for objective, standardized, and accessible methods to assess technical skill and provide formative feedback.[1-3]

In response to these challenges, simulation-based training (SBT) has become a cornerstone of competency-based medical education, providing safe environments where skills can be developed that transfer to real clinical settings. However, practice alone is insufficient for mastery; the limited availability of expert faculty to provide timely feedback creates a significant "feedback gap." AI represents the next technological frontier in closing this gap, with the potential to deliver objective, scalable, and personalized feedback. However, AI is not without risk—algorithms can perpetuate biases from training data, and the opacity of complex models may erode learner trust. Furthermore, simulation struggles to replicate the full context of live operations, limiting its ability to teach complex clinical judgment. Therefore, as this field rapidly expands, systematic evaluation of AI feedback systems is imperative.[4,5]

The application of AI in surgical education is a rapidly expanding field, with an increase in new systems and validation studies. This rapid growth requires a systematic evaluation of the current landscape to combine existing evidence, identify key trends, and chart a course for future development. Therefore, the goal of this study was to evaluate the role of AI in surgical feedback by conducting a comprehensive scoping review of the peer-reviewed literature. This review aims to map the characteristics of existing AI feedback systems, analyze the evidence for their efficacy, and identify critical research themes, evidence gaps, and priorities for future investigation.

METHODS

Methodological Framework

This scoping review was designed and conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines.[6] The methodology was established in advance to ensure a systematic and comprehensive mapping of the existing literature.

Search Strategy and Sources

A systematic search of major academic databases, including PubMed, Embase, and Web of Science, was conducted on May 7, 2025.The search was limited to literature published between January 2015 and May 2025. This timeframe was selected to capture the period of significant advancement in deep learning and AI applications in medicine, beginning with the widespread adoption of convolutional neural networks following key developments in 2012 to 2014, while allowing sufficient time for educational applications to emerge in the literature. The search strategy, developed in consultation with a medical librarian, combined keywords and subject headings across 3 core concepts: (1) artificial intelligence (e.g., “artificial intelligence,” “machine learning,” “deep learning,” “neural network”), (2) surgical education (e.g., “surgical education,” “medical education,” “simulation,” “training,” “assessment”), and (3) feedback (e.g., “feedback,” “coaching,” “guidance”).

Study Selection and Eligibility Criteria

The study selection process was managed using Covidence, a web-based software platform for systematic review management. The initial search yielded 2374 unique studies. Following the removal of duplicates, a 2-stage screening process was conducted. First, 2 independent reviewers screened the titles and abstracts of all retrieved articles. The initial screening of 1813 studies was completed to identify potentially relevant papers. Subsequently, the full texts of these articles were retrieved and assessed for final eligibility by the 2 reviewers. Disagreements at either stage were resolved through discussion and consensus with a third senior reviewer.

Studies were included if they met the following criteria: (1) described the development or implementation of an AI-based system; (2) the system’s purpose was to provide feedback or assessment on a surgical or procedural task; and (3) the feedback was directed at a surgical learner (e.g., medical student, resident, fellow, or practicing surgeon). Studies were excluded if they were review articles, editorials, or if they described AI systems for applications other than educational feedback (e.g., diagnostic assistance, robotic control). This rigorous process resulted in the inclusion of 18 studies for the final synthesis.

Data Extraction and Synthesis

A structured data extraction form was created to systematically collect key information from each of the 18 included studies. The extracted variables included: first author and year of publication, surgical specialty and specific procedure, study design, characteristics of the AI technology and algorithms used, feedback modality (real-time vs. post-task), primary and secondary outcomes, the training environment (simulation vs. real surgery), and participant demographics (number and level of expertise). The extracted data were then synthesized

through a qualitative thematic analysis to identify, analyze, and report recurring patterns and core concepts across the body of literature.

RESULTS

Overview of Included Studies

The study selection process is summarized in the PRISMA flow diagram (Fig. 1). The 18 included studies were published between 2018 and 2025. A risk of bias analysis, using the ROBINS-I tool,[7] revealed that twelve studies (67%) presented a moderate risk of bias. The research covered a breadth of specialties, with 6 studies (33%) in general surgery, 3 (17%) in neurosurgery, 2 (11%) in urology, 2 (11%) in interventional pulmonology, and one each in otolaryngology, plastic surgery, gastroenterology, and neonatology. A variety of study designs were employed, including 5 (28%) randomized controlled trials (RCTs).

Applications and Technologies

The included studies reported on AI systems applied across a wide range of surgical domains and training environments (Table 1). The underlying technologies also varied significantly. Foundational systems relied on established machine learning algorithms like support vector machines (SVMs) for skill classification in VR simulation.[4] More recent systems leveraged deep learning models such as convolutional neural networks (CNNs) for analyzing surgical video[8] and recurrent neural networks (RNNs) for sequential motion data.[9] Some research applies natural language processing (NLP) techniques to automatically categorize the raw transcripts of verbal feedback given by human surgeons, discovering clinically meaningful themes without pre-labeled data.[6]

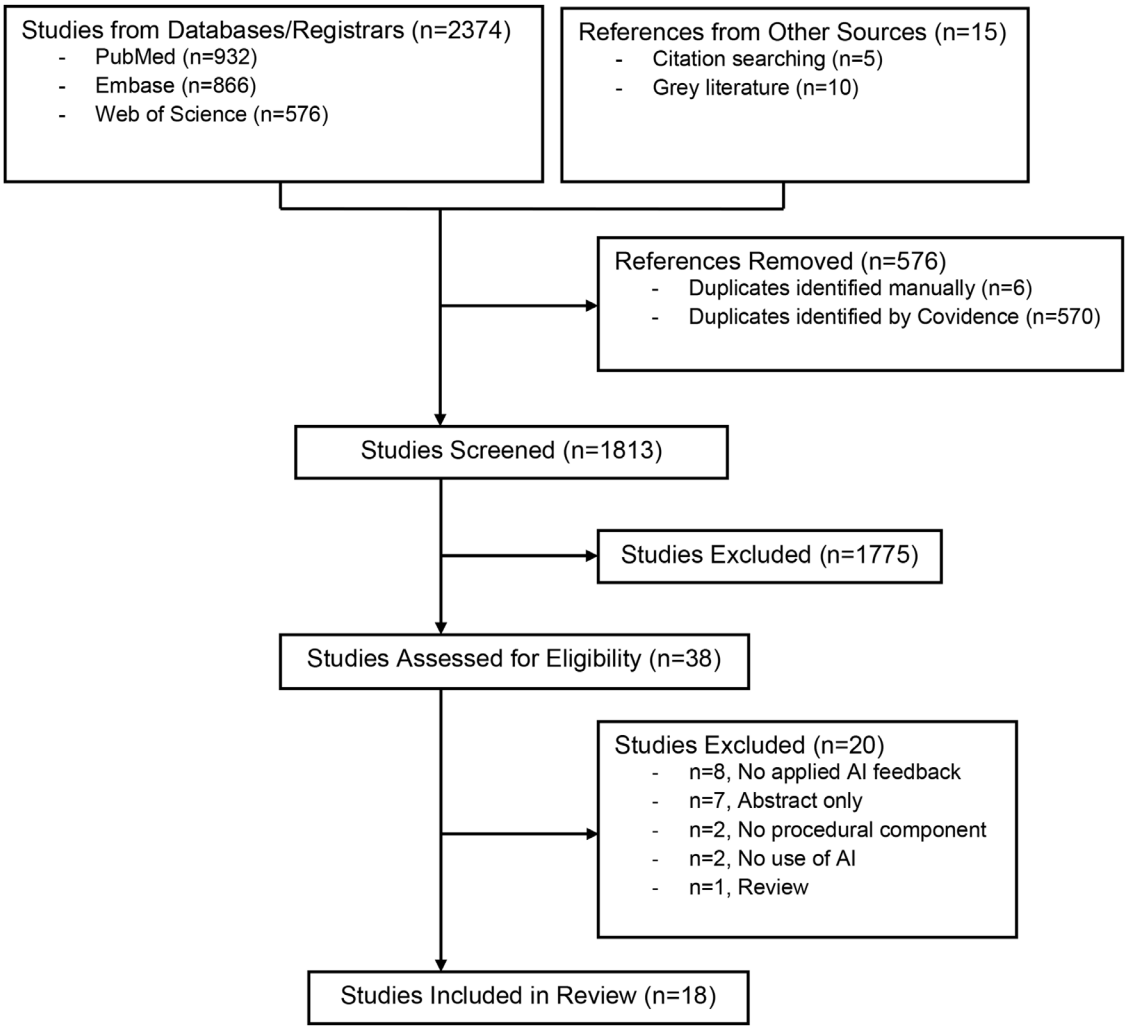

FIGURE 1. PRISMA diagram.

TABLE 1. Characteristics of Included Studies

Study	Surgical Specialty/ Procedure	Study Design	AI Technology/ Algorithm	Feedback Modality	Key Outcomes	Setting	Participants (N, Level)
Sheth et al.[13]	Pediatrics/neonatal thoracentesis	Feasibility study	Siamese convolutional neural network (SCNN)	Real-time & post-task	Skill assessment scores, user satisfaction	Simulation (hybrid)	5 (fellows, neonatologist)
St John et al.[3]	General surgery/ laparoscopic cholecystectomy (LC)	Quasi-experimental	Deep neural networks (GoNo-GoNet)	Post-task (gamified)	Accuracy scores, user perception	Simulation (mobile App)	903 (students, trainees, surgeons)
Yibulayimu et al.[11]	Plastic surgery/ liposuction	Validation study	Random forest, XGBoost, SHAP (XAI)	Real-time & post-task	Classification accuracy (expert vs. novice)	Simulation & real surgery	12 (experts, novices)
Yilmaz et al.[1]	Neurosurgery/ brain tumor resection	RCT	Long short-term memory (LSTM) networks (ICEMS)	Real-time & post-task	Composite performance score, OSATS, cognitive load	Simulation (VR)	97 (medical students)
Kocielnik et al.[10]	Urology/robot-assisted surgery	Validation study	Unsupervised topic modeling (BERTopic), NLP	Post-task (analysis of human feedback)	Prediction of trainee behavioral change	Real surgery	N/A (analysis of 3740 feedback instances)
Kiyasseh et al.[2]	Urology/RARP suturing	Validation study	Vision transformer, attention mechanism, TWIX	Post-task (explanations)	Reliability of explanations (AUPRC), explanation bias	Real surgery	100 + (surgeons, students) from 3 hospitals
Fazlollahi et al.[12]	Neurosurgery/ brain tumor resection	Cohort study (secondary analysis of RCT)	Support vector machine (SVM)	Post-task	Unintended performance outcomes (safety vs. efficiency)	Simulation (VR)	46 (medical students), 14 (experts)
Noroozi et al.[14]	General surgery/ LC	Feasibility study (Beta test)	Deep learning (Go/No-Go zones)	Post-task (gamified)	User scores vs. experience level, user satisfaction	Simulation (mobile app)	29 (students, trainees, surgeons)
Ma et al.[15]	Urology/robotic suturing	RCT (Pilot)	AI-based video analysis	Post-task	EASE scores, targeted improvement for underperformers	Simulation (dry-lab)	42 (novices)
Wu et al.[8]	General surgery/ LC	RCT	CNN-based system (SurgSmart)	Post-task	LCRF scores, CVS achievement	Real surgery	18 (novice surgeons)
Mirchi et al.[4]	Neurosurgery/ brain tumor resection	Validation study	Support vector machine (SVM)	Post-task	Classification accuracy (skilled vs. novice)	Simulation (VR)	50 (students, residents, experts)
Cold et al.[18]	Pulmonology/ bronchoscopy	RCT	AI-based segment identification	Real-time & post-task	Diagnostic completeness, procedure time, structured progress	Simulation (phantom)	20 (novices)

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TABLE 1 (*continued*)

Study	Surgical Specialty/ Procedure	Study Design	AI Technology/ Algorithm	Feedback Modality	Key Outcomes	Setting	Participants (N, Level)
Holden et al.[16]	Interventional radiology/US-guided needle insertion	Validation study	Decision trees, fuzzy inference systems	Post-task	GRS score prediction, feedback usefulness	Simulation (phantom)	24 (trainees, experts)
Wijewickrema et al.[19]	Otolaryngology/ temporal bone surgery	Comparative study	Rule-based vs. neural network	Real-time	Checklist & global scores (CS & GS)	Simulation (VR)	26 (medical students)
Yao et al.[9]	Gastroenterology/ endoscopy	Pre-post trial	Deep convolutional neural networks (DCNN)	Post-task	ADR, GPC detection rate	Real procedures	16 (endoscopists)
Trujillo et al.[17]	General surgery/ LC	Cross-sectional	Large language model (ChatGPT 4.0)	Post-task	Feedback usefulness & quality, Error detection rate	Real surgery (text scenarios)	4 (residents), 3 (surgeons), 2 (experts)
Agbontaen et al.[20]	Critical care/ bronchoscopy	RCT	AI navigation with augmented reality	Real-time	Mean intersegmental time, procedure time, cognitive load	Simulation (phantom)	40 (critical-care physicians)
Fazlollahi et al.[5]	Neurosurgery/ brain tumor resection	RCT	SVM (VOA), LSTM (ICEMS)	Post-task	ICEMS expertise score, OSATS	Simulation (VR)	70 (medical students)

Abbreviations: ADR, adenoma detection rate; AUPRC, area under the precision-recall curve; CS, checklist score; CVS, critical view of safety; EASE, end-to-end assessment of suturing expertise; GPC, gastric precancerous conditions; GRS, global rating scale; GS, global score; ICEMS, intelligent continuous expertise monitoring system; LCRF, laparoscopic cholecystectomy rating form; NLP, natural language processing; OSATS, objective structured assessment of technical skills; RCT, randomized controlled trial; RARP, robot-assisted radical prostatectomy; TWIX, training with explanations; VOA, virtual operative assistant; VR, virtual reality; XAI, explainable AI.

Another study evaluated the ability of a large language model (LLM), ChatGPT 4.0, to provide useful feedback based on text descriptions of surgical scenarios.[10] Data inputs for these systems included instrument kinematics (position, orientation, velocity), computer vision analysis of surgical video (instrument tracking, event detection), and transcribed verbal communications.

Pedagogical Design and Efficacy

Fourteen of the eighteen studies (78%) described post-task feedback systems, where feedback was delivered after the procedure through performance dashboards, automatically generated reports, or annotated video replays.[2-6,8-17] Four studies (22%) detailed real-time feedback systems that provided immediate, intra-procedural guidance via auditory instructions or visual overlays on the surgical scene.[1,18-20]

Five RCTs directly compared AI feedback to human expert instruction or a no-feedback control group.[1,5,8,18,20] In a trial by Yilmaz et al.[1] real-time AI feedback in VR neurosurgery training resulted in the AI group achieving significantly better performance outcomes than both the expert-instructed and control groups. Similarly, an RCT comparing AI-guided bronchoscopy training to human expert instruction concluded that the AI group demonstrated significantly faster and more efficient performance.[20] Wu et al.[8] showed that an AI-assisted coaching program for live laparoscopic cholecystectomy, which provided post-operative video analysis, led to significant improvements in surgical performance and safety compared to a self-learning control group. A subgroup analysis by Ma et al.[15] in the context of robotic suturing training found that while all participants benefited from practice, the AI-based video feedback provided a significantly larger improvement for participants identified as “underperformers” at baseline.

Advanced Concepts Reported in the Literature

Several studies addressed the challenge of opaque AI models by focusing on explainable AI (XAI). These systems were designed to provide transparent reasoning for their assessments. For example, some systems used techniques to create visual summaries that highlighted which specific features contributed most to a skill assessment,[11] while others generated heatmaps over surgical video frames to indicate which moments were most critical to the AI’s evaluation.[4] One study investigated algorithmic fairness, identifying an “explanation bias” where the reliability of AI-generated explanations differed across surgeon sub-cohorts.[2]

A secondary analysis of an RCT by Fazlollahi et al.[12] reported the existence of a “hidden curriculum.” While an AI tutor focused on safety and instrument movement successfully improved performance on those metrics (e.g., reducing damage to healthy tissue), it also had negative, unintended effects on unassessed metrics, as trainees in the AI group became significantly slower and had a lower rate of tumor removal. The impact of AI feedback on the learner’s cognitive state was also explored, with conflicting findings. One study reported that their real-time auditory AI feedback system resulted in a significantly higher extraneous cognitive load for learners compared to in-person human instruction.[1] In contrast, another study using the NASA Task Load Index found no significant difference in overall perceived workload between their AI-guided group and an expert tutor group in bronchoscopy training.[20]

DISCUSSION

Principal Findings: The Maturation of an Educational Technology

This scoping review charts the rapid maturation of AI as an educational technology in surgery. The principal finding is that AI-powered feedback is rapidly moving from a futuristic concept to a promising educational intervention with the potential to produce tangible improvements in procedural skill acquisition. The evidence from several key studies suggests that these systems can accelerate learning curves, leading to performance that is not only superior to unassisted practice but often on par with, or even exceeding, the outcomes achieved with direct instruction from human experts.[1,20] However, as this review finds, the current body of evidence is not yet mature. With a majority of studies (67%) presenting a moderate risk of bias and validation quality in the wider literature described as "suboptimal," these findings must be interpreted as evidence of potential, not of widespread, validated efficacy. By offering scalable, standardized, and objective assessments, AI tutors can provide learners with consistent, high-quality feedback that is essential for deliberate practice and skill mastery.

This evolution can be conceptualized as an integrated, closed-loop system for skill acquisition. This process begins with data capture from a procedure, whether it be instrument motion, force application, or video, which is fed into an AI engine for analysis against expert benchmarks. The engine generates targeted feedback, which is delivered to the trainee, who then integrates it into their subsequent performance, restarting the cycle. However, a deeper analysis reveals that the landscape of AI feedback is not monolithic, and the design of an effective AI tutor involves complex pedagogical choices, not merely technical ones. The most critical challenge identified is the "hidden curriculum," as demonstrated by Fazlollahi et al.[12] Their AI tutor, the Virtual Operative Assistant

(VOA), by optimizing for safety metrics, successfully reduced damage to healthy tissue but at the cost of significantly slower surgical performance. This suggests that an AI, in its current form, acts as a technical optimizer, not a clinical mentor. It excels at assessing discrete, measurable, technical tasks, but struggles to understand or teach surgical judgment (the high-order cognitive skill of balancing competing, ambiguous priorities like safety and efficiency under pressure).

Similarly, there is a trade-off between minimizing cognitive load and maximizing information delivery. Real-time auditory feedback may provide constant correction but, as Yilmaz et al.[1] showed, can increase the extraneous cognitive load on the learner, suggesting that processing a continuous stream of instructions required greater mental effort. In contrast, Agbontaen et al.[20] found no difference in perceived workload between their AI-guided and expert-tutor groups, suggesting the impact on cognitive load is highly dependent on the interface, modality, and frequency of the feedback. Finally, a trade-off exists between the immediacy of real-time feedback, which corrects errors instantly but may foster dependence, and the reflective nature of post-task feedback, which allows errors to persist during practice but may lead to deeper, more consolidated learning.[1,5]

This finding directly addresses a fundamental gap in the current literature. The systems in this review were almost exclusively tested on well-defined, fundamental procedural tasks in controlled simulation environments, often with medical student participants rather than surgical trainees. None were found to assess performance in ambiguous clinical scenarios or high-stakes emergencies, which are defined by their unpredictability. This suggests that an AI tutor, if deployed naively, may inadvertently teach trainees to optimize for a score at the expense of developing the adaptive, real-world judgment required in a live operation.[21]

This highlights the profound importance of not just "careful curriculum design," but of active human expert oversight. The solution to the "hidden curriculum" is not simply a better algorithm; it is a human-in-the-loop collaborative model.[22] The goal should be a partnership, where AI provides the objective, granular data, and the human educator provides the context, wisdom, and judgment. This reframes the central research question from a simple “Does AI work?” to a more sophisticated “How do we design AI systems to achieve specific, and sometimes competing, educational goals?”

Implications for Surgical Curriculum Integration

The findings of this review carry significant implications for the future of surgical training. AI tutors have the potential to act as powerful democratizing agents, providing all trainees, regardless of their institution’s size, location, or faculty resources with access to standardized, objective, and expert-level feedback. The development of mobile gaming apps like LapBot, for example, makes sophisticated training tools accessible on a personal device, removing the barrier of expensive, centralized simulation centers.[3,14] This can help level the playing field and reduce the variation in training quality that currently exists. Furthermore, by providing objective, granular data on performance, these systems can enable a true transition to proficiency-based progression, where advancement is tied to the demonstrated mastery of skills rather than the time spent in a rotation.

The cautionary findings on unintended consequences and the hidden curriculum are critically important.[12] The integration of AI into surgical curricula must be a thoughtful, deliberate process. Educators cannot simply “plug in” an AI tutor and expect optimal outcomes. Instead, curricula must be designed to complement the AI, ensuring that trainees develop a holistic skill set. This might involve using AI for foundational technical skills while reserving human faculty time for higher-order cognitive skills like strategic planning, intraoperative judgment, and managing unexpected complications. The goal should be a collaborative partnership, where AI provides the data and objective assessment, and the human educator provides the context, wisdom, and mentorship. The finding that AI feedback may be particularly beneficial for “underperforming” learners also suggests a role for AI in targeted, personalized remediation, allowing instructors to identify and support struggling trainees more effectively.[15]

Moreover, the emergence of explainable AI is a critical step toward building trust and improving pedagogical value.[4,11] By moving away from “black box” models to systems that can show their reasoning. For example, by highlighting which specific features or video frames contributed to an assessment, AI feedback becomes more transparent and actionable for the learner. However, as AI systems become more integrated into high-stakes decision-making, ensuring their fairness is paramount. The identification of “explanation bias,” where AI feedback is not equally reliable for all surgeons, is a crucial finding that highlights the need for rigorous validation and bias mitigation strategies to ensure that all trainees receive feedback of comparable quality.[2]

Future Research Directions

Despite the rapid progress, this review identifies several critical gaps in the literature. A primary concern is the potential for unintended consequences, such as the

"hidden curriculum" that may create a trade-off between safety and efficiency. The impact of AI on cognitive load also requires further investigation, as current evidence is conflicting. Foundational issues of explainability and trust remain critical, as overcoming the "black box" problem is necessary to ensure feedback is understood and accepted. Concurrently, researchers must address algorithmic and explanation bias to guarantee fair and equitable feedback for all trainees. From a validation standpoint, there is a significant lack of long-term data, particularly concerning skill retention and the transfer of skills to the live operating room. The cost-effectiveness of implementing AI tutors has yet to be formally analyzed.

First, there is a significant lack of long-term data. The vast majority of studies are single-session interventions. While they demonstrate impressive short-term skill acquisition, it remains unknown whether these skills are retained over weeks or months. This addresses the critical, and largely unanswered, question of skill transferability. Future research must prioritize longitudinal studies to determine not only if skills are retained, but if they successfully transfer from the simulator to the live operating room. Furthermore, as noted in the discussion, a significant gap exists in applying these AI tutors to "ambiguous clinical scenarios." Current models focus on teaching defined, technical skills. Research is urgently needed to explore AI's role in assessing and teaching the higher-order cognitive skills essential for surgery: complex decision-making, team communication, and managing unexpected intraoperative complications.

Second, the cost-effectiveness of these systems has not been formally evaluated. The development, validation, and implementation of high-fidelity simulators and sophisticated AI algorithms requires significant resources. Rigorous economic analyses are needed to weigh the upfront costs against the potential long-term benefits of accelerated training, reduced complication rates, and increased operating room efficiency.

Third, there is an opportunity for the development of multimodal AI systems that integrate a wider array of data. Current systems typically rely on 1 or 2 data streams, such as video or kinematics. Future systems could create a more comprehensive model of surgical expertise by integrating a wider array of data, including haptic force profiles, instrument energy usage, transcribed verbal communication,[6] and even the surgeon's physiological data to assess cognitive state. Furthermore, there are opportunities to expand these AI feedback systems into other complex specialties, such as otolaryngology-head and neck surgery, building on existing work in areas like virtual reality temporal bone surgery to provide nuanced feedback on microsurgical techniques.[19]

Finally, the field should move towards exploring models of human-AI collaboration. Research using AI to analyze and categorize the feedback given by human surgeons represents a major shift from AI as a replacement for the instructor to AI as an augmentation tool.[6] Future research should investigate how AI can serve as a "co-pilot" for human educators, providing them with real-time data and insights to make their own feedback more targeted and effective.

Limitations

This scoping review has several limitations. The search was restricted to a specific timeframe and a selection of databases, and some relevant studies may have been missed. The variation in AI technologies, procedures, and outcome measures across the included studies makes direct meta-analytic comparison challenging. Additionally, the generalizability of findings is limited by several factors. The majority of included studies utilized medical students rather than surgical residents as participants, and many of the assessed procedures, such as bronchoscopy and ultrasound-guided needle insertion, represent foundational procedural skills rather than complex surgical operations. Furthermore, the included literature spans multiple specialties, including interventional pulmonology and gastroenterology, which are not strictly surgical disciplines. Therefore, while the evidence suggests AI can enhance procedural skill acquisition in simulation, caution is warranted in generalizing these findings to technically demanding surgical procedures or advanced surgical trainees.

CONCLUSION

This scoping review provides a comprehensive map of the current landscape of AI for feedback in surgical education. The evidence clearly demonstrates that AI has evolved from an emerging concept into a promising and significant educational tool. Several rigorous studies have indicated that AI-driven feedback can accelerate the acquisition of technical skills, in some cases outperforming traditional teaching methods. However, the maturation of the field is also revealing a new set of sophisticated challenges. Issues of explain ability, algorithmic fairness, cognitive load, and the potential for unintended pedagogical consequences are now at the forefront of the research agenda. The future of surgical education will not be a choice between human or machine, but rather a collaborative partnership between them. To realize this future, the research community must continue its commitment to rigorous, evidence-based investigation that prioritizes not only technological innovation but also sound pedagogical design.

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